

Scaling analysis for boundary layer equations

BL EQ ($\delta \ll L$) $\leftarrow$ This means that velocity field is unaffected by the solid plate except for the thin layer

[Derivation of BL EQs from Navier Stokes EQs]

Scaling analysis for a steady 2-dimensional flow over flat plate

- Mass conservation

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow \frac{u_\infty}{L} \sim \frac{v_0}{\delta} \Rightarrow v_0 \sim \frac{L}{\delta} u_\infty$$

$$\therefore u \gg v$$

(u_∞, v_0 : characteristic velocities for u and v)

- x -MTM

$$\begin{aligned} u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \\ &\Downarrow \\ \sim \frac{u_\infty^2}{L} &\sim v_0 \frac{u_\infty}{\delta} \sim \frac{\Delta p}{\rho L} \sim \nu \frac{u_\infty}{L^2} \sim \nu \frac{u_\infty}{\delta^2} \end{aligned}$$

- y -MTM

$$\begin{aligned} u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \\ &\Downarrow \\ \sim u_\infty \frac{v_0}{L} &\sim \frac{v_0^2}{\delta} \sim \frac{\Delta p}{\rho \delta} \sim \nu \frac{v_0}{L^2} \sim \nu \frac{v_0}{\delta^2} \\ \frac{u_\infty^2}{L} \left(\frac{\delta}{L} \right) &\frac{u_\infty^2}{L} \left(\frac{\delta}{L} \right) \frac{\Delta p}{\rho \delta} \nu \frac{u_\infty}{L^2} \left(\frac{\delta}{L} \right) \nu \frac{u_\infty}{\delta^2} \left(\frac{\delta}{L} \right) \end{aligned}$$

Comparison of x -momentum equation with y -momentum equation

$\rightarrow$ Inertial force and viscous force terms in y -momentum equation are small

$$\rightarrow \frac{\partial p}{\partial y} = 0 \rightarrow p = p(x) \rightarrow \frac{\partial p}{\partial x} = \frac{dp}{dx} = \frac{dp}{dx} \Big|_{y=any} = \frac{dp}{dx} \Big|_{y=\infty} = \frac{dp_\infty}{dx} = 0$$

(Note that pressure gradient vanishes as $y \rightarrow \infty$ because velocity is constant outside BL)

This means the pressure gradient term can be neglected in the x -momentum equation.

Consequently, the mass and momentum conservation equations are reduced to

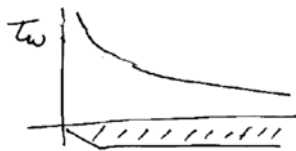
$$\begin{aligned} \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} &= 0 & u(x, y=0) &= v(x, 0) = 0 \\ u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} &= \nu \frac{\partial^2 u}{\partial y^2} & u(x, y=\infty) &= u_\infty \end{aligned}$$

(Note)

- 3 equations (3 unknowns u, v, p) in original NS EQs $\rightarrow$ 2 equations (2 unknowns u, v) in BL EQs
- The above equations can be solved analytically by the similarity solution method (Blasius solution)

[Results by scaling analysis]

- The scaling analysis also yields $\frac{\delta}{L} \sim \sqrt{\frac{\nu}{u_\infty L}} = \sqrt{\frac{1}{\text{Re}_L}}$ $\frac{\delta}{x} \sim \sqrt{\frac{1}{\text{Re}_x}} = \text{Re}_x^{-\frac{1}{2}}$
 - High speed, small $\nu \rightarrow$ Thin BL
 - Shear stress by scaling analysis
- $$\tau_x = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} \sim \mu \frac{u_\infty}{\delta} \sim \sqrt{\frac{\rho \mu u_\infty}{x}} \mu u_\infty \Rightarrow C_{f,x} \sim \text{Re}_x^{-\frac{1}{2}}$$
- Thin BL $\rightarrow$ Large velocity gradient $\rightarrow$ Large shear stress
 - $\tau_x \sim x^{-1/2}$ ($\because$ BL thickness $\sim x^{1/2}$)



- Shear stress proportional to $u_\infty^{3/2}$ (Why? $\tau_x \sim \mu \frac{u_\infty}{\delta}$)
- ...

(Note)

- In transient heat conduction, analysis of the diffusion equation $\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$ shows that thermal (temperature) penetration over a time interval t is $\sim \sqrt{\alpha t}$
 - Similarly, momentum diffusion length over a time interval t is $\sim \sqrt{\nu t}$
 - BL thickness $\sim$ length over which velocity (momentum) changes significantly
- momentum $\sim$ diffusion length $\Rightarrow \delta \sim \sqrt{\nu t} \sim \sqrt{\nu \frac{L}{u_\infty}}$